

CS 6762
Fall 2023
SP, ML, FC
Homework 2

Answer all questions. Work alone. Total 100 points. Type or write neatly.

1. Describe the two required and one typical property of a linear system. (5 points)
2. Assume the following signal which is described by a 2-tuple where the first number is time and the second number is the value of temperature. Draw the impulse decomposition for this signal. (0,0) (1,80) (2, 85) (3, 90) (4,87) (5, 88). (5 points)
3. Using convolution compute the output signal $y[n]$ when the input signal is $x[n]$ and the system impulse response signal $h[n]$ are both given as below; again the 2-tuple is (time, value of signal). (10 points)

$x[n] = (0,0) (1, -1) (2,-1.5) (3,2) (4,1.5) (5,1.5) (6, 0.8) (7,0) (8,-.5)$

$h[n] = (0,1) (1,-.8) (2,-.5) (3,-.2)$

4. Suppose that you have an input signal called XX in the time domain of values 1,1,2,1,3,2 for times 0 through 5 , respectively. The algorithm below can compute the DFT for this signal. Change the code below to compute the DFT for this signal. Show a trace of the program execution, for just lines 310 through 380, when K=1. (20 points)

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100 'THE DISCRETE FOURIER TRANSFORM
110 'The frequency domain signals, held in REX[ ] and IMX[ ], are calculated from
120 'the time domain signal, held in XX[ ].
130 '
140 DIM XX[511]           'XX[ ] holds the time domain signal
150 DIM REX[256]          'REX[ ] holds the real part of the frequency domain
160 DIM IMX[256]          'IMX[ ] holds the imaginary part of the frequency domain
170 '
180 PI = 3.14159265        'Set the constant, PI
190 N% = 512               'N% is the number of points in XX[ ]
200 '
210 GOSUB XXXX             'Mythical subroutine to load data into XX[ ]
220 '
230 '
240 FOR K% = 0 TO 256      'Zero REX[ ] & IMX[ ] so they can be used as accumulators
250   REX[K%] = 0
260   IMX[K%] = 0
270 NEXT K%
280 '
290 '                     'Correlate XX[ ] with the cosine and sine waves, Eq. 8-4
300 '
310 FOR K% = 0 TO 256      'K% loops through each sample in REX[ ] and IMX[ ]
320   FOR I% = 0 TO 511    'I% loops through each sample in XX[ ]
330     '
340     REX[K%] = REX[K%] + XX[I%] * COS(2*PI*K%*I%/N%)
350     IMX[K%] = IMX[K%] - XX[I%] * SIN(2*PI*K%*I%/N%)
360     '
370   NEXT I%
380 NEXT K%
390 '
400 END

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TABLE 8-2

5. Explain the difference between feature extraction and feature matching. (5 points)
6. What is the frequency response of a system? (5 points)
7. What is the FFT, where and when should it be used, and what is its complexity? (5points)
8. Physical world signals may contain amplitude or frequency information as the primary aspects of the signals. For each of the following indicate in which domain the most important information lies and explain: (i) number of pedestrians that walked through a door; (ii) detecting the word “log-off”; (iii) monitoring the presence absence of a vehicle on a quiet country road; and (iv) discriminating between a car and a bus. (10 points)
9. Describe what a spectrogram is (discussed in class and in text as well in Chapter 22). (5 points)

10. What is the purpose of a Hamming window? (5 points)
11. Identify 2 instances where you would need to use a DFT in a smart city when doing signal processing. (5 points)
12. Identify 2 signals for autonomous systems where you might use either convolution or DFT depending on the specific objective. Be sure to state the objective in each case. (10 points)
13. Identify 2 places where you would use a convolution for smart health applications and describe why. (10 points)